

Joint Optimisation of Load Balancing and Handover for Hybrid LiFi and WiFi Networks

Xiping Wu, Majid Safari and Harald Haas

Li-Fi Research and Development Centre, Institute for Digital Communications

School of Engineering, The University of Edinburgh

EH9 3JL, Edinburgh, UK

{xiping.wu, majid.safari, h.haas}@ed.ac.uk

Abstract—Recently a promising concept of hybrid networks based on light fidelity (LiFi) and wireless fidelity (WiFi) emerged. The idea is to combine ultra-small cell LiFi networks with ubiquitous coverage radio frequency (RF) communication systems. In such a hybrid network, WiFi access points (APs) serve a relatively large coverage area with limited bandwidth, and are thus susceptible to traffic overload. This issue is magnified with an increasing number of users because of the inefficient medium access control (MAC) in WiFi systems. LiFi can alleviate this issue by providing additional capacity. LiFi cells, however, have a limited coverage and this could result in significant handover overhead. A conventional load balancing (LB) method optimises the network throughput when the signal-to-noise ratio (SNR) of each user is known and fixed. Although this method delivers maximum throughput at a given time instance, it fails to consider the throughput loss due to handover, especially in an indoor scenario where users may frequently switch between APs. Taking the handover overhead into account, in this paper we propose a novel LB method that focuses on optimising the network throughput over a period of time. Simulation results show that the proposed method can increase the system throughput by up to 70% compared to existing LB methods.

Index Terms—LiFi, WiFi, attocell, hybrid network, handover, load balancing

I. INTRODUCTION

MOBILE data traffic has been exponentially increasing and continues to grow. The Cisco visual networking index (VNI) forecast that global mobile data traffic will increase eightfold between 2015 and 2020, reaching 30.6 exabytes per month by 2020 [1]. During the same period, average spectral efficiency is predicted to increase by only 50%, leading to a looming spectrum crunch. To meet the future demand for mobile data traffic, new wireless communication technologies in the mm-wave region of the RF spectrum are introduced [2]. A natural extension is to consider the nm-region of the electromagnetic spectrum, especially as there are inexpensive electronic devices with powerful communication properties. This has led to the introduction of visible light communication (VLC) [3]. In VLC intensity modulation is applied to light-emitting diodes (LEDs) to convey information. Photodiodes (PDs), semiconductor devices that convert light into current, are employed at the receiver to detect signals, and this is referred to as direct detection. Compared with traditional radio-frequency (RF) system, VLC exhibits many advantages: i) it is a cost effective solution as it utilises

existing lighting infrastructures to deliver unprecedented data rates; ii) it benefits from a vast unregulated spectrum of up to 300 THz (1000 times the entire RF spectrum); iii) it offers high security as light does not propagate through walls; and iv) it can be used safely in RF sensitive environments, such as aircraft cabins, petrochemical plants, etc. Recent research shows a single VLC link with a single micro-LED is capable of achieving data rates as high as 3 Gbps [4].

The wireless networking extension of point-to-point VLC is referred to as LiFi [5], where a typical AP covers an ultra-small optical attocell with a diameter of approximately 2-3 m [5]. LiFi and WiFi networks do not suffer from mutual interference [6]. It was shown that a hybrid WiFi/LiFi network can significantly improve the performance of indoor wireless communications, and that the net performance gain is more than the sum performance of individual stand-alone networks [7].

However, AP selection (APS) and mobility management become the most challenging issues facing a hybrid network. Compared with LiFi APs, WiFi APs have relatively large coverage areas but limited system capacities. Consequently, using signal strength strategy (SSS), which is commonly used in a homogeneous network, would make the WiFi system susceptible to traffic overload. A cooperative load balancing (LB) scheme was proposed in [8], aiming to maximise the throughput of a hybrid network for a given time instance. We refer to this method as instant LB (ILB), which may lead to frequent handovers for mobile users. Therefore, when the handover overhead is taken into account, ILB does not guarantee optimum performance over a period of time. There are two types of handovers in a hybrid network: horizontal handover (HHO) between homogeneous APs and vertical handover (VHO) between heterogeneous APs. In general, VHO takes a longer time to complete than HHO. In [9], a dynamic LB method was introduced, where the coverage area of each AP is modified according to the handover overhead. However, this method may still incur a substantial number of VHOs, which can significantly compromise network throughput.

In this paper, the mobility management issue of a hybrid LiFi and WiFi network is formulated as a joint optimisation problem of LB and handover. During the period of interest, each user is assigned to a specific network and is then handed over between homogeneous APs when needed. Simulation

results show that the proposed method can greatly improve the network throughput compared to ILB, especially when users move relatively fast.

The remainder of this paper is organised as follows. In Section II, the system model of an indoor hybrid LiFi and WiFi network is described, including the network deployment and channel model. The novel LB method is proposed in Section III. Monte Carlo simulation results are presented in Section IV. Finally, the conclusion is drawn in Section V.

II. SYSTEM MODEL

A. Hybrid LiFi and WiFi Network

We consider a hybrid LiFi/WiFi indoor wireless network. As shown in Fig. 1, a number of LiFi APs and one WiFi AP are installed in a square-shaped room. Each LiFi AP is operated jointly with a ceiling LED lamp facing vertically downwards, while the WiFi AP is placed on the ground and in the room centre. Note that a LiFi AP covers a confined area, while the WiFi AP provides coverage for the entire room, and beyond. LiFi APs are assumed to operate at different frequencies, and thus they do not interfere with each other. Roaming users are assumed. They randomly change directions and velocities. Each user can be connected to one AP only, either a LiFi AP or the WiFi AP. For each AP, time-division multiple access (TDMA) is considered as the multiuser access technique.

B. LiFi Channel Model

As shown in Fig. 2, a LiFi channel is comprised of two components: line-of-sight (LoS) and non line-of-sight (NLoS) paths. Given AP i and user u , the LoS path is the straight line between them, and the corresponding Euclidean distance is denoted by $d_{i,u}$. The angles of irradiance and incidence are denoted by $\phi_{i,u}$ and $\psi_{i,u}$, respectively. The channel gain of the LoS path is represented by $H_{\text{LoS}}^{i,u}$, and it is formulated as [10, Eq. 10]:

$$H_{\text{LoS}}^{i,u} = \frac{(m+1)A_{\text{pd}}}{2\pi d_{i,u}^2} \cos^m(\phi_{i,u}) g_f g_c(\psi_{i,u}) \cos(\psi_{i,u}), \quad (1)$$

where $m = -\ln 2 / \ln(\cos \Phi_{1/2})$ is the Lambertian emission order, and $\Phi_{1/2}$ is the radiation angle at which the intensity is half of the intensity at the main-beam direction; A_{pd} denotes the physical area of PD; g_f is the gain of the optical filter; $g_c(\psi_{i,u})$ is the optical concentrator gain:

$$g_c(\psi_{i,u}) = \begin{cases} \frac{n^2}{\sin^2(\Psi_{\max})}, & 0 \leq \psi_{i,u} \leq \Psi_{\max} \\ 0, & \psi_{i,u} > \Psi_{\max} \end{cases}, \quad (2)$$

where n denotes the refractive index, and $\Psi_{\max}$ is the semi-angle of the field of view (FoV) of the PD.

For the NLoS components, it has been shown that it is generally sufficient to only take the first-order reflections into account [11]. A first-order reflection consists of two segments: a) from AP i to a point w on the wall, and b) from the point w to user u . The Euclidean distances of these two segments are respectively denoted by $d_{i,w}$ and $d_{w,u}$. The angles of radiance and incidence with respect to the first segment are $\phi_{i,w}$ and

$\vartheta_{i,w}$, and for the second segment they are $\vartheta_{w,u}$ and $\psi_{w,u}$. The channel gain contributed by NLoS is expressed as (3), where A_w stands for a small area on the wall and ρ_w denotes the wall reflectivity. Adding (1) to (3), the total gain of a LiFi channel is computed by:

$$H_{\text{LiFi}}^{i,u} = H_{\text{LoS}}^{i,u} + H_{\text{NLoS}}^{i,u}. \quad (3)$$

The photons at the receiver are converted into an electrical current:

$$I_{\text{elec}} = R_{\text{pd}} H_{\text{LiFi}}^{i,u} P_{\text{opt}} / \kappa, \quad (4)$$

where R_{pd} denotes the detector responsivity; P_{opt} is the transmitted optical power; and κ is the optical to electric power conversion coefficient. The signal-to-noise ratio (SNR) of a LiFi user is written as:

$$\gamma_{\text{LiFi}}^{i,u} = \frac{(R_{\text{pd}} H_{\text{LiFi}}^{i,u} P_{\text{opt}} / \kappa)^2}{N_{\text{LiFi}} B_{\text{LiFi}}}, \quad (5)$$

where N_{LiFi} is the power spectral density (PSD) of noise at the PD, including shot noise and thermal noise, and B_{LiFi} is the system bandwidth of a LiFi AP.

C. WiFi Channel Model

Although a single WiFi AP is considered, the index of AP is retained for the purpose of generality. The gain of a WiFi channel is denoted by $G_{\text{WiFi}}^{i,u}$, and it is given by [12]:

$$G_{\text{WiFi}}^{i,u} = \left| H_{\text{WiFi}}^{i,u} \right|^2 10^{\frac{-L(d_{i,u}) + X_\sigma}{10}}, \quad (6)$$

where $H_{\text{WiFi}}^{i,u}$ describes the channel transfer function whose magnitude is assumed to follow a Rayleigh distribution¹; the shadow fading X_σ is a zero-mean Gaussian random variable with a standard deviation of 10 dB; and $L(\cdot)$ denotes the free-space path loss:

$$L(d) = \begin{cases} 20 \log_{10}(f_c d) - 147.5, & d < d_{\text{ref}} \\ 20 \log_{10} \left(f_c \frac{d^{2.75}}{d_{\text{ref}}^{1.75}} \right) - 147.5, & d \geq d_{\text{ref}} \end{cases}, \quad (7)$$

where f_c is the central carrier frequency, and $d_{\text{ref}} = 10$ m is the reference distance. The SNR of a WiFi user is calculated by:

$$\gamma_{\text{WiFi}}^{i,u} = \frac{G_{\text{WiFi}}^{i,u} P_{\text{WiFi}}}{N_{\text{WiFi}} B_{\text{WiFi}}}, \quad (8)$$

where N_{WiFi} is the PSD of noise at the receiver; P_{WiFi} and B_{WiFi} are the transmit power and bandwidth consumed by a WiFi AP, respectively.

¹We appreciate that a more precise model would be to consider a Rician channel until a certain breakpoint distance. The simplified model, however, is justified as it does not change the general trend of the results, and a more accurate WiFi channel model will be considered in a journal extension.

$$H_{\text{NLoS}}^{i,u} = \int_{A_w} \frac{(m+1)A_{\text{pd}}}{2(\pi d_{i,w}d_{w,u})^2} \rho_w \cos^m(\phi_{i,w}) g_f g_c(\psi_{w,u}) \cos(\psi_{w,u}) \cos(\vartheta_{i,w}) \cos(\vartheta_{w,u}) dA_w. \quad (3)$$

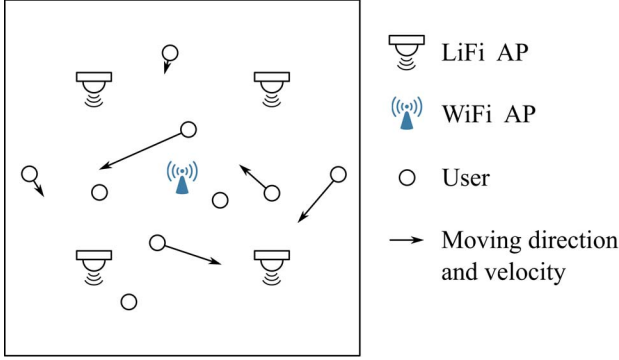


Fig. 1. Paradigm of an indoor hybrid LiFi and WiFi network.

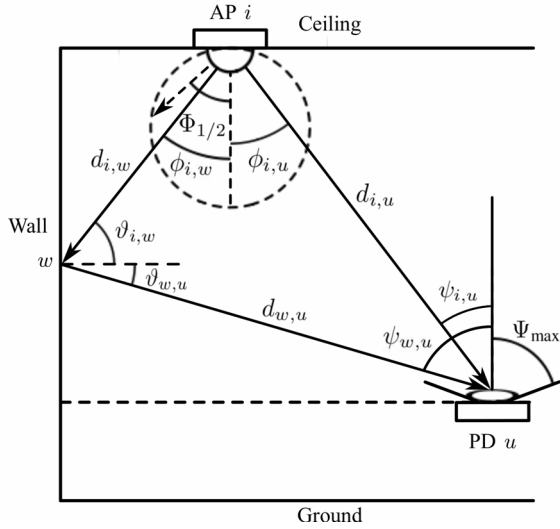


Fig. 2. The LoS and first-order NLoS paths of a LiFi channel.

III. JOINT OPTIMISATION OF LOAD BALANCING AND HANDOVER

We consider a proportional-fair scheduler [13]:

$$\text{maximise } \sum_u \log(R_u), \quad (9)$$

where R_u is the data rate achieved by user u . Given a time instance t , the ILB method in [8] solves the above optimisation problem based on the instant data rate of each user, which is denoted by $R_u^{(t)}$ and computed by:

$$R_u^{(t)} = \sum_i \chi_{i,u}^{(t)} \rho_{i,u}^{(t)} r_{i,u}^{(t)}, \quad (10)$$

where $\chi_{i,u}^{(t)} = 1$ indicates there is a connection between AP i and user u at time instance t , while $\chi_{i,u}^{(t)} = 0$ means otherwise;

$\rho_{i,u}^{(t)}$ is a fraction variable between 0 and 1, which stands for the proportion of time that AP i is allocated to user u ; $r_{i,u}^{(t)}$ is the Shannon capacity that AP i can provide to user u :

$$r_{i,u}^{(t)} = B_i \log_2 \left(1 + \gamma_{i,u}^{(t)} \right). \quad (11)$$

Note that each user can be assigned to one AP only, and therefore $\sum_i \chi_{i,u}^{(t)}$ in (10) can be moved out of the logarithmic function in (9). Substituting (10) into (9), the objective function (OF) of ILB is obtained by:

$$\Gamma_{\text{ILB}} \left(\chi^{(t)}, \rho^{(t)} \right) = \sum_u \sum_i \chi_{i,u}^{(t)} \log \left(\rho_{i,u}^{(t)} r_{i,u}^{(t)} \right), \quad (12)$$

where $\chi^{(t)}$ and $\rho^{(t)}$ are the sets of $\chi_{i,u}^{(t)}$ and $\rho_{i,u}^{(t)}$ for all i and u , respectively. The optimisation problem of ILB is then formulated as:

$$\begin{aligned} & \text{maximise} \quad \Gamma_{\text{ILB}} \left(\chi^{(t)}, \rho^{(t)} \right) \\ & \text{subject to} \quad \chi_{i,u}^{(t)} \in \{0, 1\}; \\ & \quad \sum_i \chi_{i,u}^{(t)} = 1; \\ & \quad 0 \leq \rho_{i,u}^{(t)} \leq 1; \\ & \quad \sum_u \chi_{i,u}^{(t)} \rho_{i,u}^{(t)} \leq 1. \end{aligned} \quad (13)$$

Unlike ILB providing the optimal solution at a given time instance, our method focuses on a period of time, during which users may require access to different APs. Therefore instead of a specific AP, the proposed method aims to determine the network that each user should be connected to. After the network type is selected, each user is served by its primary AP, which is defined as the AP having the highest SNR in the selected network. Let κ denote the network type, which is either LiFi or WiFi. For user u being assigned to network κ , the proportion of time consumed by the handover process is denoted by $\lambda_{\kappa,u}$, and it is calculated by:

$$\lambda_{\kappa,u} = \begin{cases} \frac{T_{\text{HHO}}^{\kappa}}{T_{\kappa,u}} & \text{if } T_{\text{HHO}}^{\kappa} \leq T_{\kappa,u} \\ 1 & \text{if } T_{\text{HHO}}^{\kappa} > T_{\kappa,u} \end{cases}, \quad (14)$$

where T_{HHO}^{κ} is the time required by one handover between two κ -type APs, and $T_{\kappa,u}$ denotes the cell dwell time (CDT) of user u , which can be statistically collected [14]. In the considered indoor scenario only a single WiFi AP is used. Therefore, there is no handover in the WiFi system, i.e. $\lambda_{\text{WiFi},u} = 0$. With respect to those users connected to LiFi, they are handed over between LiFi APs when necessary. The average data rate achieved by user u is computed by:

$$\bar{R}_u = \sum_{\kappa} \chi_{\kappa,u} r_{\kappa,u} \min\{\rho_{\kappa,u}, 1 - \lambda_{\kappa,u}\}, \quad (15)$$

where $\chi_{\kappa,u}$ and $\rho_{\kappa,u}$ are similar to $\chi_{i,u}$ and $\rho_{i,u}$, but related to the network type rather than a specific AP; the operator $\min\{\cdot\}$ returns the minimum value of a set of numbers; and $r_{\kappa,u}$ denotes the average Shannon capacity that a κ -type AP can provide to user u . Like CDT, $r_{\kappa,u}$ can be measured statistically as follows:

$$r_{\kappa,u} = \sum_{i \in \kappa} \chi_{i,u} \int_t B_i \log_2(1 + \gamma_{i,u}^{(t)}) dt. \quad (16)$$

The coefficient $\min\{\rho_{\kappa,u}, 1 - \lambda_{\kappa,u}\}$ in (15) signifies the proportion of time that is available for data transmission. If $T_{\kappa,u}$ is infinite, i.e. the user is stationary, then $\lambda_{\kappa,u} = 0$ and $\min\{\rho_{\kappa,u}, 1 - \lambda_{\kappa,u}\} = \rho_{\kappa,u}$. On the contrary, for a fast-moving user, $1 - \lambda_{\kappa,u}$ is very small and this limits the available time for data transmission. Substituting (15) into (9), the OF of the proposed method is given by:

$$\Gamma(\chi, \rho) = \sum_u \sum_{\kappa} \chi_{\kappa,u} \log(r_{\kappa,u} \min\{\rho_{\kappa,u}, 1 - \lambda_{\kappa,u}\}). \quad (17)$$

The optimisation problem of the proposed method is then formulated as:

$$\begin{aligned} & \text{maximise} && \Gamma(\chi, \rho) \\ & \text{subject to} && \chi_{\kappa,u} \in \{0, 1\}; \\ & && \sum_{\kappa} \chi_{\kappa,u} = 1; \\ & && 0 \leq \rho_{\kappa,u} \leq 1; \\ & && \sum_u \chi_{\kappa,u} \rho_{\kappa,u} \leq N_{\text{AP}}^{\kappa}, \end{aligned} \quad (18)$$

where N_{AP}^{κ} is the number of κ -type APs.

IV. SIMULATION RESULTS

In this section, Monte Carlo simulations are conducted to evaluate the performance of the proposed method against ILB and SSS. Consider a square-shaped room measuring 5 m on side, which accommodates four LiFi APs and one WiFi AP. Also, the random waypoint model [15] is adopted to mimic individuals' movement behaviour. The user's velocity is a random variable uniformly distributed between 0 and $v_{\max}$, and is assumed constant during the period of interest. The additional parameters required for the simulations are summarised in Table I. Take $v_{\max} = 2$ m/s and 5 m/s for example, the occurrence rates of HHO and VHO are shown in Fig. 3. The proposed method fully suppresses VHO, as expected. Meanwhile, it nearly doubles the occurrence rate of HHO in ILB. Nevertheless, the occurrence rate of HHO when using the new method is far lower than that of VHO in ILB. This trend becomes more pronounced as $v_{\max}$ increases. Also, HHO incurs a lower overhead cost than VHO. Therefore, compared to ILB, the proposed method can significantly reduce the overhead cost caused by handover. With respect to SSS, the occurrence rates of HHO and VHO are both lower than in ILB. But, SSS still incurs a much higher handover overhead cost than the proposed method. Fig. 4 presents the system throughput as a function of the number of users in the room. The ideal situation that neglects the overhead cost of handover

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Room size	$5 \times 5 \times 3$ m
The physical area of a PD, A_{pd}	1 cm^2
The gain of the optical filter, g_f	1
The refractive index, n	1.5
Half-intensity radiation angle, $\Phi_{1/2}$	60°
FoV semi-angle of PD, $\Psi_{\max}$	90°
Optical to electric power conversion coefficient, κ	3
Detector responsivity, R_{pd}	0.53 A/W
Wall reflectivity, ρ_w	0.8
Transmitted optical power per LiFi AP, P_{opt}	3 Watt
Transmitted power per WiFi AP, P_{WiFi}	20 dBm
Bandwidth per LiFi AP, B_{LiFi}	20 MHz
Bandwidth per WiFi AP, B_{WiFi}	20 MHz
PSD of noise in LiFi, N_{LiFi}	$10^{-21} \text{ A}^2/\text{Hz}$
PSD of noise in WiFi, N_{WiFi}	-174 dBm/Hz
The overhead cost of HHO, T_{HHO}	200 ms
The overhead cost of VHO, T_{VHO}	500 ms

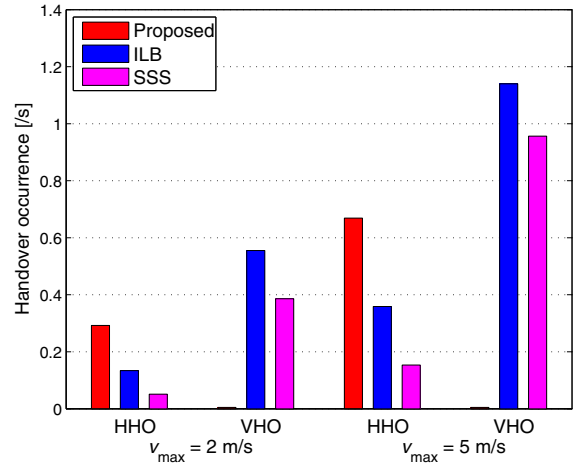


Fig. 3. The occurrence rates of HHO and VHO.

is also taken into account for each method. Three outcomes are observed. First, ILB is optimal in the ideal situation as it is designed to be, but it dramatically falls behind the proposed method when handover costs are considered. Second, SSS always performs worst among the three methods. This is because although SSS yields a smaller number of handovers than ILB, it can cause traffic congestions especially to WiFi, and this compromises the system throughput significantly. Third, for various numbers of users, the proposed method achieves a stable increase of roughly 50% in throughput against ILB. When $N_u = 2$, for example, the proposed method obtains a throughput of 364 Mbps, which is 52.6% more than the 238 Mbps of ILB. As N_u increases to 14, the throughput of the proposed method is 785 Mbps, creating an increase of 53.6% over the 511 Mbps of ILB.

In Fig. 5, the system throughput is shown as a function of the user's velocity, while the number of users is fixed to

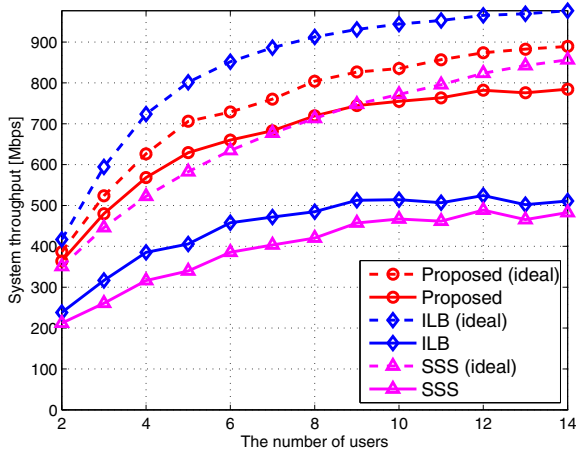


Fig. 4. System throughput against N_u , while $v_{\max}$ is set at 5 m/s.

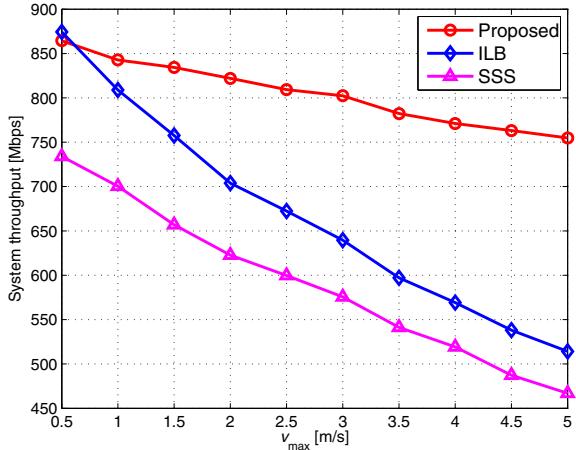


Fig. 5. System throughput against $v_{\max}$, while N_u is set at 10.

10. As shown, only when users are moving very slowly, the proposed method is surpassed by ILB with a quite modest gap. At $v_{\max} = 0.5$ m/s, the throughput of the proposed method is merely 1% less than that of ILB. After $v_{\max}$ exceeds 0.6 m/s, the proposed method keeps outperforming ILB. In addition, as $v_{\max}$ increases, the system throughput becomes lower for all three methods. However, the throughput of the proposed method reduces more slowly than that of ILB and SSS. Therefore the gain achieved by the proposed method increases as $v_{\max}$ increases. With every 1 m/s increase in $v_{\max}$, the throughput gap between the proposed method and ILB increases by about 10%.

V. CONCLUSION

In this paper, a novel method was proposed to cope with the mobility management issue in hybrid LiFi/WiFi networks. The approach works on the principle that users that would trigger frequent handover in the LiFi system are assigned to the WiFi AP. Based on this underlying principle, the proposed method

performs joint optimisation of load balancing and handover. Unlike in ILB which selects a specific AP for each user at every time instance, the new method determines the most suitable network a user is served by during a given period. Simulation results validate that the proposed method can significantly reduce the occurrence rate of handover compared to ILB. Specifically, compared to ILB and SSS, the proposed method improves the system throughput by up to 50% and 70%, respectively. Future research will focus on analysing the optimality and computational complexity of the proposed method.

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